

# Graduate Seminar 2

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# Graduate Seminar 2

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Section 1 - Solving for $\psi(x, t)$

- We solve  $\psi(x, t)$  by a numerical resolution of the Time-Dependent Schrödinger Equation (TDSE), i.e

$$i \frac{\partial}{\partial t} \psi(x, t) = \left( \hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \quad \psi(x, 0) = \psi_0(x), \quad (1)$$

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x. \quad (2)$$

$$E_0 = -\kappa^2/2, \quad \psi_0(x) = \kappa^{1/2} e^{-\kappa|x|}, \quad \kappa = 1, \quad (3)$$

$$E_k = k^2/2, \quad \psi_k^{(\pm)}(x) = e^{ikx} + R(\pm|k|)e^{\pm i|kx|}, \quad R(k) = \frac{i\kappa}{k - i\kappa}. \quad (4)$$

- The parameters of the laser we will be using are given by the relations

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t'), \quad 0 \leq t \leq T, \quad t' = t - T/2, \quad (5)$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad n_{oc} = 5. \quad (6)$$

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## Density, survival probability and PEMD

- The density probability of the wavefunction, at  $t$  is given by

$$\mathcal{D}(x, t) = |\psi(x, t)|^2 = \int \psi^*(x, t) \psi(x, t) dx. \quad (7)$$

- The coefficients of the projection of  $\psi$  are then, at  $t$ , and by definition, expressed as

$$a_0 = e^{iE_0 t} \int \psi_0(x) \psi(x, t) dx, \quad a_k = e^{iE_k t} \int \psi_k^{(-)*}(x) \psi(x, t) dx. \quad (8)$$

- Then,

$$P_0 = |a_0|^2, \quad \text{and} \quad P(k) = |a_k|^2. \quad (9)$$

- The ionization probability can then be found many different ways that should be consistent with each others

$$P_{\text{ion}} = \int P(k) \frac{dk}{2\pi} = 1 - P_0. \quad (10)$$

- $P(k)$  is the probability for the electron being ionized with the energy  $E_k$ .

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## Observables

- For an arbitrary operator  $\mathcal{O}$ , we have

$$\mathcal{O}(t) = \int \psi^*(x, t) \mathcal{O} \psi(x, t) dx \quad (11)$$

- The the Fourier transform is obtained using

$$\mathcal{O}(\omega) = \int \mathcal{O}(t) e^{i\omega t} dt. \quad (12)$$

- Following Bransden and Joachain, "Quantum Mechanics (2000)" (see 11.145), we define the transition amplitude  $\mathcal{O}_{ba}$  and probability  $P_{ba}$  between two states  $a$  and  $b$  such as

$$\mathcal{O}_{ba} = \int dx \psi_b^*(x) \mathcal{O} \psi_a(x), \quad P_{ba} = |\mathcal{O}_{ba}|^2. \quad (13)$$

When continuum states between  $q$  and  $q + dq$  are involved, the quantity of interest is

$$P_{ba} dq = \left| \int dx \psi_b^*(x) \mathcal{O} \psi_a(x) \right|^2 dq. \quad (14)$$

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## Length, velocity and acceleration

- We want to evaluate the quantities

$$d(t) = -\langle \hat{x} \rangle = -\int \psi^*(x, t) \hat{x} \psi(x, t) dx, \quad \text{and} \quad \dot{x}(t) = \frac{d}{dt} x(t), \quad (15)$$

$$p(t) = \langle \hat{p} \rangle = -i \int \psi^*(x, t) \frac{\partial}{\partial x} \psi(x, t) dx. \quad (16)$$

- Direct numerical differentiation can sometimes be avoided with the help of the Ehrenfest theorem and the commutator algebra

$$[\hat{p}^2, \hat{x}] = -2i\hat{p} \implies \hat{p} = i[\hat{H}, \hat{x}], \quad (17)$$

$$\frac{d}{dt} \langle \hat{x} \rangle = \langle \hat{p} \rangle, \quad \frac{d}{dt} \langle \hat{p} \rangle = \left\langle -\frac{\partial V_{\text{ext}}}{\partial x} \right\rangle. \quad (18)$$

and some others (see Bransden and Joachain, Quantum Mechanics (2000), p.531-532)

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## High Harmonic Generation (HHG) - (semi-classical approx.)

- In the classical approx., the spectrum is obtained from the following (Eq.(80))

$$Q_{\nu}^{(c)}(\omega) = \alpha^2 \omega^2 |\mathbf{A}_{\nu} \mathbf{d}(\omega)|^2, \quad (19)$$

Let us apply the following simplifications

$$\nu \longrightarrow \omega, \quad \mathbf{A}_{\nu} = \sqrt{\frac{2\pi c^2}{\omega L^3}} \mathbf{e}_{\nu} \longrightarrow A_{\omega}, \quad \text{and} \quad \alpha A_{\omega} = 1. \quad (20)$$

- So the observable we are considering here is

$$Q^{(c)}(\omega) = \omega^2 |d(\omega)|^2. \quad (21)$$

Additionally let us consider Eq. (104)

$$Q^{(c)}(\omega) = |p(\omega)|^2. \quad (22)$$

and, by virtue of  $\mathbf{p}(t) = -\dot{\mathbf{d}}(t)$ , which reads

$$Q^{(c)}(\omega) = \left| -\dot{\mathbf{d}}(\omega) \right|^2. \quad (23)$$

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## Additional note from Ehrenfest theorem

- Using the integration by parts formula, we get

$$p(\omega) = \int_{-T/2}^{T/2} e^{i\omega t} \frac{dx(t)}{dt} dt, \quad u(t) = x(t), \quad v(t) = e^{i\omega t}, \quad (24)$$

thus

$$p(\omega) = \int_{-T/2}^{T/2} \frac{d}{dt} \left( e^{i\omega t} x(t) \right) dt - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt = \left[ e^{i\omega t} x(t) \right]_{-T/2}^{T/2} - i\omega \int_{-T/2}^{T/2} e^{i\omega t} x(t) dt, \quad (25)$$

which reduces to (21) up to a constant.



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## Transition amplitude

- From Yangaliev *et al.*, we have

$$p_{k0} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_0(x) = \frac{2k\mathcal{K}^{3/2}}{\mathcal{K}^2 + k^2}, \quad (26)$$

$$p_{kk'} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_{k'}^{(-)}(x) = 2\pi k \delta(k - k') - \frac{2\mathcal{K}}{k^2 - k'^2 + i\epsilon} \left( \frac{k'|k|}{|k| - i\mathcal{K}} - \frac{k|k'|}{|k'| + i\mathcal{K}} \right). \quad (27)$$

- Using (17), we can rewrite those expressions as

$$p_{k0} = -i(E_k - E_0) d_{k0}, \quad (28)$$

$$p_{kk'} = -i(E_k - E_{k'}) d_{kk'}. \quad (29)$$

- Those quantities are another way to study convergence, at the structural level

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Part 2 - Solving $\phi_\omega(x, t)$

- We are now interested in solving the following 1D problem with a ZRP

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left( \hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left( \hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} S(x, t) \end{cases} \quad (30)$$

where

$$\hat{H}_a = -\frac{1}{2} \frac{d^2}{dx^2} - \kappa \delta(x), \quad \hat{V}_{\text{ext}}(t) = F(t)x, \quad S(x, t) = \hat{p} \psi(x, t) \quad (31)$$

$$\phi_\omega(x, 0) = \frac{-i\kappa^{3/2}}{\omega} \text{sgn}(x) \left( e^{-\kappa|x|} - e^{-\kappa(\omega)|x|} \right), \quad \psi(x, 0) = \kappa^{1/2} e^{-\kappa|x|}, \quad (32)$$

$$\hat{p} = -id/dx, \quad \text{and} \quad \kappa(\omega) = \sqrt{\kappa^2 + 2\omega}. \quad (33)$$

$$\phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t_0) - i \int_{t_0}^t U(t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

# High-order exponential splitting (composition methods) (Phys. Rev. A 101, 013410 (2020))

## Yoshida (1990) - 4th-order symmetric decomposition

$$\phi_{\omega}(x, t + \Delta t) = U_4(t + \Delta t, t) \phi_{\omega}(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- From the following Strang operator  $U(h) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t+h/2)h} e^{-iE_n h/2}$ , we compose

$$U_4(h) = U(c_1 h) U(c_2 h) U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \quad (35)$$

- More precisely, we can observe the following

**Theorem 8.6** (Hairer, Solving ODE II p.554)

If  $\Psi_h$  represent a one-step method that is symmetric, of order  $p = 2k$  and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \quad 2c_1^{p+1} + c_2^{p+1} = 0, \quad (36)$$

imply that the composition (35) is of order  $p + 2$ .

If you start from the Strang method, then in the above  $k = 1$ .

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## Midpoint Duhamel - 2nd order approximation

$$\phi_{\omega}(x, t + \Delta t) = U(t + \Delta t, t)\phi_{\omega}(x, t) - i \int_t^{t+\Delta t} U(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- We can show that the above can be expressed as

$$\phi_{n+1}^{\omega} = U(\Delta t)\phi_n^{\omega} - i\Delta t e^{i\omega t_{n+1/2}} S_{n+1/2}, \quad (37)$$

where the source is evaluated at  $t_{n+1/2} = t + \Delta t/2$ , i.e and  $S_{n+1/2} = U(\Delta t/2)S_n$ , with  $S_n = S(x, t_n)$

- This is the midpoint Duhamel formula, of order 2.
- It's what will allow us to get familiar with the numerical intricacies of our problem
- The driver shall be regarded as decent if the expected order is met

# Geometric propagators

## Simpson Duhamel - 4th-order quadrature

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(x, \tau) d\tau. \quad (34)$$

- The Simpson quadrature consists in writing

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left( U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (38)$$

where the source is evaluated at  $t_n$ ,  $t_{n+\frac{1}{2}}$  and  $t_{n+1}$

- This is the Simpson Duhamel formula, of order 4.
- It requires a 4th order evolution operator to work
- Simpson-Duhamel multiplies the number of operations required by Midpoint-Duhamel by 3.

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## Observables

$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx, \quad (39)$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx, \quad (40)$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (41)$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi} \quad (42)$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi} \quad (43)$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Midpoint Duhamel - Self conv. test results for  $S(x, t) = \hat{p}\psi(x, t)$  - ZRP

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} \tag{38}$$

| $t$        | $\Re \psi(1)$  | $\Im \psi(1)$  | $\ \psi\ $ | $\Re \phi(2)$  | $\Im \phi(2)$  | $\ \phi\ $ |
|------------|----------------|----------------|------------|----------------|----------------|------------|
| -1102.3132 | 1.0000000      | 0.0000000      | 1.0000000  | 0.0000000      | 1.4733146(-4)  | 0.6074294  |
| -1080.2669 | 1.0000000      | -1.5299150(-5) | 1.0000000  | 9.0161878(-8)  | 1.4733146(-4)  | 2.9977570  |
| -1058.2207 | 1.0000000      | -3.0598299(-5) | 1.0000000  | 1.8032375(-7)  | 1.4733145(-4)  | 4.2519438  |
|            |                |                | ...        |                |                |            |
| -22.0463   | -4.6878735(-7) | 1.1266426(-6)  | 1.0000000  | -3.4864304(-3) | -2.0577179(-3) | 0.7712524  |
| -0.0000    | 4.8643339(-1)  | -9.9881227(-1) | 1.0000000  | -1.3514523     | -8.0617898(-4) | 3.3039060  |
| 22.0463    | 2.8495297(-7)  | 1.2366110(-6)  | 1.0000000  | -3.7992762(-3) | 1.3930728(-3)  | 4.2371529  |
|            |                |                | ...        |                |                |            |
| 1058.2207  | -9.9526450(-1) | -9.7202282(-2) | 1.0000000  | 1.5396368(-5)  | -1.4659389(-4) | 1.1029687  |
| 1080.2669  | -9.9526599(-1) | -9.7187055(-2) | 1.0000000  | 1.5389349(-5)  | -1.4659255(-4) | 3.5763755  |
| 1102.3132  | -9.9526748(-1) | -9.7171829(-2) | 1.0000000  | 1.5382713(-5)  | -1.4659538(-4) | 4.1762517  |

| Level          | $\Delta t$ (relative) | $E_\psi$      | $E_\phi$      | $P_\psi/P_\phi$ |
|----------------|-----------------------|---------------|---------------|-----------------|
| 1              | $\Delta t_0$          | 0.1185025(-1) | 0.1632911(+0) | —               |
| 2              | $\Delta t_0/2$        | 0.2953778(-2) | 0.3951342(-1) | 2.004 / 2.048   |
| 3              | $\Delta t_0/4$        | 0.7378959(-3) | 0.9798974(-2) | 2.001 / 2.011   |
| Observed order |                       |               |               | 2.0011/2.0116   |

Ne = 100, Np = 10  
-xmin = xrange/2 = Np Ns/2  
 $\tau = 2\pi n_{oc}/\omega_0$ ,  $T = 8\tau$

Initial conditions and parameters

$$S(x, t) = -i\partial_x \psi(x, t)$$
$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t), \quad t' = t - T/2$$
$$F_0 = 0.03, \quad \omega_0 = 0.114, \quad \omega = 0.5$$
$$\Delta t_0 = 2.204, \quad n_{oc} = 5$$

■ Results of the time propagation for the zero range potential with and without source

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Simpson Duhamel - Self conv. test results for $S(x, t) = \hat{p}\psi(x, t)$ - ZRP

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left( U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (38)$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad (35)$$

| $t$        | $\Re \psi(1)$  | $\Im \psi(1)$  | $\ \psi\ $ | $\Re \phi(2)$  | $\Im \phi(2)$  | $\ \phi\ $ |
|------------|----------------|----------------|------------|----------------|----------------|------------|
| -1102.3132 | 1.0000000      | 0.0000000      | 1.0000000  | 0.0000000      | 1.4733146(-4)  | 0.6074294  |
| -1080.2669 | 1.0000000      | -1.5299150(-5) | 1.0000000  | 9.0161878(-8)  | 1.4733146(-4)  | 2.9977570  |
| -1058.2207 | 1.0000000      | -3.0598299(-5) | 1.0000000  | 1.8032375(-7)  | 1.4733145(-4)  | 4.2519438  |
|            |                |                | ...        |                |                |            |
| -22.0463   | -4.6878735(-7) | 1.1266426(-6)  | 1.0000000  | -3.4864304(-3) | -2.0577179(-3) | 0.7712524  |
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|            |                |                | ...        |                |                |            |
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| Level          | $\Delta t$ (relative) | $E_\psi$      | $E_\phi$      | $p_\psi/p_\phi$ |
|----------------|-----------------------|---------------|---------------|-----------------|
| 1              | $\Delta t_0$          | 0.5564105(-3) | 0.1529880(-1) | —               |
| 2              | $\Delta t_0/2$        | 0.3480484(-4) | 0.1093299(-2) | 3.999 / 3.807   |
| 3              | $\Delta t_0/4$        | 0.2175758(-5) | 0.7047939(-4) | 4.000 / 3.955   |
| Observed order |                       | 3.9997/3.9553 |               |                 |

$$Ne = 100, \quad Np = 10$$

$$-x_{\min} = x_{\text{range}}/2 = Np Ns/2$$

$$\tau = 2\pi n_{oc}/\omega_0, \quad T = 8\tau$$

Initial conditions and parameters

$$S(x, t) = -i\partial_x \psi(x, t)$$

$$F(t) = F_0 e^{-(2t'/\tau)^2} \cos(\omega_0 t), \quad t' = t - T/2$$

$$F_0 = 0.03, \quad \omega_0 = 0.114, \quad \omega = 0.5$$

$$\Delta t_0 = 2.204, \quad n_{oc} = 5$$

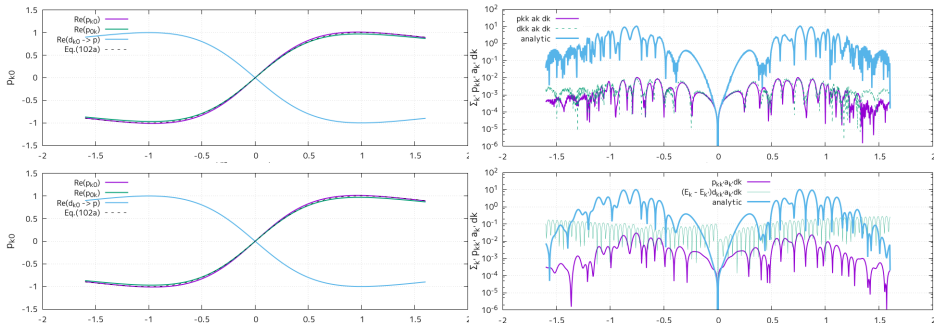
■ Results of the time propagation for the zero range potential with and without source

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$



# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Transition amplitudes - gauge-related convergence



$N_p = 20, \quad N_s = 100$   
 $x_{\text{max}} \sim N_p N_s / 2$   
 $\omega_0 = 0.114 \text{ a.u.}$   
 $F_0 = 0.03 \text{ a.u.}, \quad T = 8\tau$

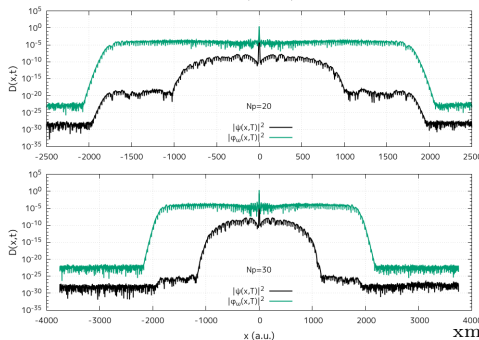
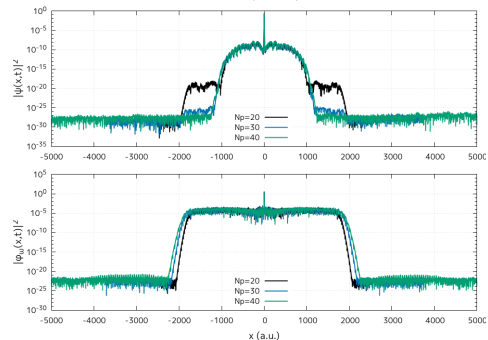
$$p_{k0} = \int dx \psi_k^{(-)*}(x) \hat{p} \psi_0(x) \quad (44)$$

$$\sum_{k'} |p_{kk'} a_k| dk = \sum_{k'} \left| \int dx \psi_k^{(-)*}(x) \hat{p} \psi_k^{(-)}(x) a_k \right| dk \quad (45)$$

U:  $N_p = 20, N_s = 100, x_{\text{max}} = 1000$   
 D:  $N_p = 40, N_s = 200, x_{\text{max}} = 4000$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Convergence region for  $\phi_\omega$  in  $x$ ,  $T = 5\tau$ , with  $F_0 = 0.03$  a.u. and  $\omega_0 = .114$  a.u.



- The convergence region for  $\psi$  is  $x_{\text{range}} \sim 2000$
- The convergence region for  $\phi$  is  $x_{\text{range}} \sim 4500$
- The plateau for  $\phi$  seems to be  $1e-8$ , above the double precision limit
- The result is confirmed by 2 different methods and has been tested against an analytical result

$$N_s = 200$$

$$x_{\text{max}} \sim (N_p N_s + q) / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

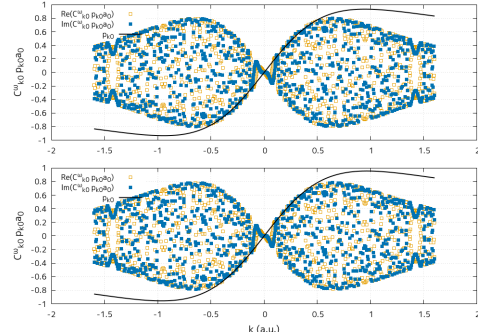
$$q = 1/4$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Convergence of the vector, non integral transition terms $p_{k0}a_0$



$$C_{k0}^\omega p_{k0}a_0 = \frac{e^{i(E_k + \omega - E_0)T}}{E_k + \omega - E_0} p_{k0}a_0$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

$$N_p = 20, \quad N_s = 200$$

$$x_{\max} \sim (N_p N_s + q) / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

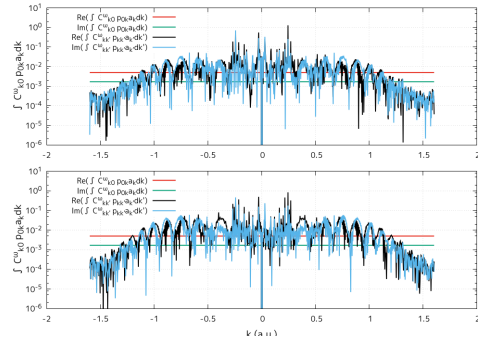
$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Convergence of the integral transition terms $p_{0k'} a_{k'}$ , and $p_{kk'} a_{k'}$



$$\int dk C_{0k}^{\omega} p_{0k} a_k = \int \frac{dk}{2\pi} \frac{e^{i(E_0 + \omega - E_k)T}}{E_0 + \omega - E_k} p_{0k} a_k$$

$$\int dk' C_{kk'}^{\omega} p_{kk'} a_{k'} = \int \frac{dk'}{2\pi} \frac{e^{i(E_k + \omega - E_{k'})T}}{E_k + \omega - E_{k'}} p_{kk'} a_{k'}$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

$$N_p = 20, \quad N_s = 200$$

$$x_{\text{max}} \sim (N_p N_s + q) / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

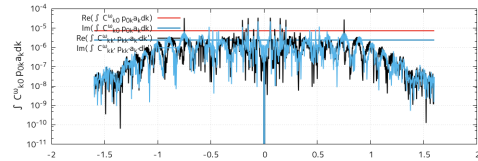
$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T}}{E_0 + \omega - E_{k'} + i0} \frac{p_{0k'} a_{k'}}{2\pi} dk'$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

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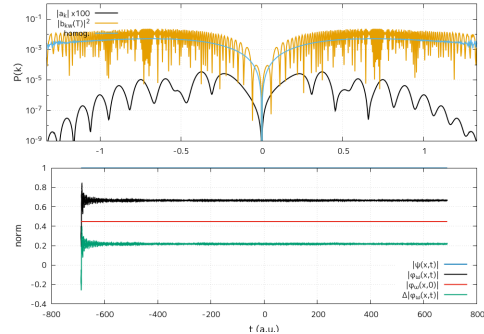
$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T}}{E_0 + \omega - E_{k'} + i0} \frac{p_{0k'} a_{k'}}{2\pi} dk'$$

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# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Study of the the coefficients $b_{k\omega}(T)$ , norm and PEMD (a look back)



$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx,$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx,$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

$$N_p = 20, \quad N_s = 200$$

$$x_{\max} \sim (N_p N_s + q) / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

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# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Transition amplitude convergence

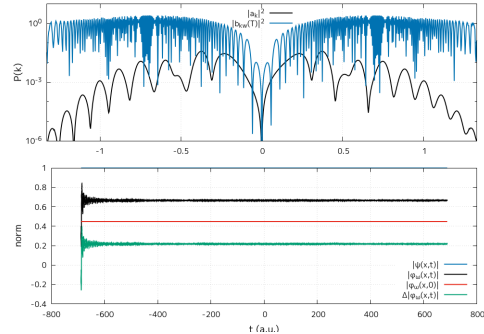
- From the two precedent slides, we can infer for  $p_{k0}$  and  $p_{kk'}$ , that
  - their associated values only depends on the structure
  - their converged values has been evaluated
  - the integral of the quantities  $p_{k0}a_0$ ,  $p_{0k}a_k$ ,  $p_{kk'}a_{k'}$  and the likes with respect to  $k$  is meant to be 0
- We can then reliably reduce the convergence of  $Q(\omega)$  to the problem of  $b_{0\omega}(T)$  and  $b_{k\omega}(T)$ 
  - **Only if we know what we are doing about those terms, that is..**
  - $\phi_\omega$  is not normed at  $t_0$  and converges toward a value a priori different from  $|\phi_\omega(x, 0)|$
  - We are yet to successfully predict said value to guide our debugging and the validity of our procedure
  - **There is an important conflict in our procedure as shown when comparing slides 20 and 21, (look at the amplitudes)**
  - **Result in 21 leads to close to converged result for  $Q(\omega)$  (previous slide), however unable to reproduce it, we have to consider 20 instead (next slide)**

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Study of the the coefficients $b_{k\omega}(T)$ , norm and PEMD (a look forward)



$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_{\omega}(x, T) dx,$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_{\omega}(x, T) dx,$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

$$N_p = 20, \quad N_s = 200$$

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$$\omega_0 = 0.114 \text{ a.u.}$$

$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$



# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Quantum High Harmonic Generation, for  $\phi_\omega$  (many channels)... - Propagation program

| $t$      | $\ \phi_{0.50}\ $ | $\ \phi_{0.51}\ $ | $\ \phi_{0.52}\ $ | $\dots$ | $\ \psi\ $ | $\ \phi_{0.50}^{(SC)}\ $ |
|----------|-------------------|-------------------|-------------------|---------|------------|--------------------------|
| —688.946 | 0.44834           | 0.44527           | 0.44226           | $\dots$ | 1.00000    | 0.44834                  |
| —688.877 | 0.40099           | 0.41982           | 0.44808           | $\dots$ | 1.00000    | 0.40099                  |
| —688.808 | 0.35620           | 0.39547           | 0.45356           | $\dots$ | 1.00000    | 0.35620                  |
| 688.739  | 0.66593           | 0.50968           | 0.46123           | $\dots$ | 1.00000    | 0.66593                  |
| 688.808  | 0.66502           | 0.50937           | 0.46113           | $\dots$ | 1.00000    | 0.66502                  |
| 688.877  | 0.66404           | 0.50905           | 0.46102           | $\dots$ | 1.00000    | 0.66404                  |

Solving the inhomogeneous equation for hundreds of channels simultaneously

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Quantum High Harmonic Generation, for  $\phi_\omega$  (many channels)... - Exploitation program

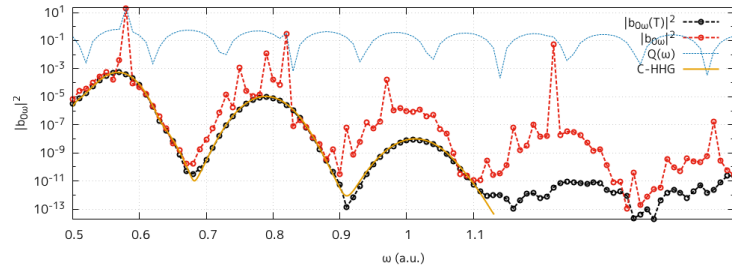
| Run         | nsteps | Re $b_0(\omega)$ | Im $b_0(\omega)$ | $Q(\omega)$     |
|-------------|--------|------------------|------------------|-----------------|
| Homogeneous | 20000  | $-0.6830301(-5)$ | $0.1928069(-5)$  | $3.2058559(-3)$ |
| Driven      | 15000  | $-0.1770619(-2)$ | $-0.3836150(-3)$ | $1.7834170(-3)$ |
| Driven      | 20000  | $-0.1770615(-2)$ | $-0.3836362(-3)$ | $1.7828446(-3)$ |

For this laser intensity and frequency, the reference result is  $Q(\omega) \sim 1e-5$

| Channel | $\omega$ (a.u.) | $Q(\omega)$     |
|---------|-----------------|-----------------|
| 1       | 0.50            | $1.9582346(-1)$ |
| 2       | 0.51            | $4.2815679(-2)$ |
| 3       | 0.52            | $2.6335454(-3)$ |
| ...     |                 |                 |
| 61      | 1.10            | $2.9983280(-1)$ |
| 62      | 1.11            | $2.1082327(-1)$ |
| 63      | 1.12            | $1.0892261(-1)$ |
| ...     |                 |                 |
| 98      | 1.47            | $7.2830156(-2)$ |
| 99      | 1.48            | $1.4370327(-1)$ |
| 100     | 1.49            | $2.0523578(-1)$ |

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Q-HHG vs C-HHG



$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx,$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx,$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

$$N_p = 20, \quad N_s = 200$$

$$x_{\max} \sim (N_p N_s + q) / 2$$

$$\omega_0 = 0.114 \text{ a.u.}$$

$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

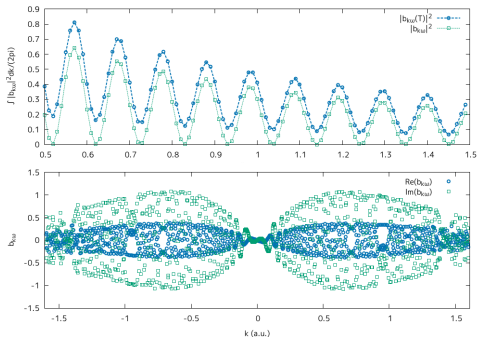
$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Study of the the coefficients  $b_{k\omega}(T)$  and  $b_{k\omega}$



$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx,$$

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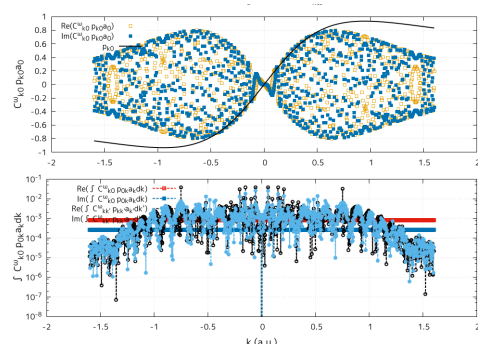
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# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Study of the structure dependent integral terms (futatabi)



$$b_{0\omega}(T) = e^{iE_0 T} \int \psi_0(x) \phi_\omega(x, T) dx,$$

$$b_{k\omega}(T) = e^{iE_k T} \int \psi_k^{(-)*}(x) \phi_\omega(x, T) dx,$$

$$Q(\omega) = |b_{0\omega}|^2 + \int |b_{k\omega}|^2 \frac{dk}{2\pi}$$

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$$q = 1/10$$

$$F_0 = 0.03 \text{ a.u.}, \quad T = 5\tau$$

$$n_{\text{steps}} = 20000, \quad \Delta t = 0.06889$$

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

# Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

## Coefficient $b_{0\omega}$ and $b_{k\omega}$ convergence

- The value of  $Q(\omega)$  for  $\omega = 0.5$  a.u is taken as a reference
  - The converged value is of the order of  $\sim 1e - 5$
  - $Q(\omega)$  is the sum of two terms, the biggest has to be of the above order as a criterion
  - $\int |b_{k\omega}|^2 dk / (2\pi)$  outweigh the required order. It indicates the direction to look for
  - $|b_{0\omega}|^2$  is likely correct ; which suggests the wavefunction  $\phi_\omega$  is
- This now looks like a convergence problem
  - likely in  $k$
  - a faster converging gauge or the analytic expression will be tested for the transition terms

$$b_{0\omega} = b_{0\omega}(T) + \int \frac{e^{i(E_0 + \omega - E_{k'})T} p_{0k'} a_{k'}}{E_0 + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$

$$b_{k\omega} = b_{k\omega}(T) + \frac{e^{i(E_k + \omega - E_0)T} p_{k0} a_0}{E_k + \omega - E_0} + \int \frac{e^{i(E_k + \omega - E_{k'})T} p_{kk'} a_{k'}}{E_k + \omega - E_{k'} + i0} \frac{dk'}{2\pi}$$